

BLOCK THE ERROR BEFORE OPENING

Ninety offline synthetic checks separate complete contracts, one-field omissions, and day-one faults. Expected decisions all match; field results remain zero.

90/90 SYNTHETIC MATCHES

72 OMISSIONS BLOCKED

6 FAULTS BLOCK OPENING

0 FIELD RESULTS

WRITE HOW COST IS MEASURED BEFORE PRICING

Six lines separate setup, recurring operation, and permanent works. Site, opening hours, and accountable buyer are unconfirmed, so planned_quantity=null; formulas are ready.

00-02 BASELINE	03-04 SETUP	05-08 RUN	09-11 REPAIR	12-13 DECIDE
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C01	FIELD BASELINE	6 ROUTES + 12 OBSERVATIONS	ONE-OFF
C02	PAPER SERVICE	PER OPEN ENTRY / COUNTER / EXIT	RECURRING
C03	MOVABLE INTERFACE	ONE SET PER AUTHORISED SITE	RENT FIRST
C04	CONTROLLED TEST	ONE INDUSTRY TEST AT A TIME	ONE-OFF
C05	STAFFED OPERATION	PER OPEN SHIFT + REPLAY SLOT	RECURRING
C06	PERMANENT WORKS	QUANTITY = 0 IN THE PILOT	FORMAL PROJECT

Rent or adapt first. Human service and independent review come before an AI display. Every installation quote includes removal, data deletion, and site restoration.

NINETY CHECKS BLOCK ERRORS BEFORE THE DOORS OPEN

This is an offline synthetic rule check, not a field result.
Complete contracts reach field review only; omissions and faults must block AI opening.

12	COMPLETE CONTRACTS	READY_FOR_FIELD_REVIEW ONLY
72	ONE-FIELD OMISSIONS	ALL_BLOCK_BEFORE_FIELD_REVIEW
6	DAY-ONE FAULT CARDS	ALL_BLOCK_AI_OPENING
90/90	EXPECTED DECISIONS MATCH	DETERMINISTIC_LOCAL_RUN
0	FIELD RESULTS	NO_ROUTE, STAFF, DEVICE, OR SERVICE PROOF

The rules block these synthetic errors. Real routes, named staff, and independent review must still answer whether the service works on site.

THREE SCALES, EACH ENDING IN A HUMAN STEP

Research connects industry and regional resources; overall design organizes walking, public space, and service; key areas turn decisions into reversible prototypes.

RESEARCH
Industry and regional loops

KEY AREAS
Reversible spatial operations

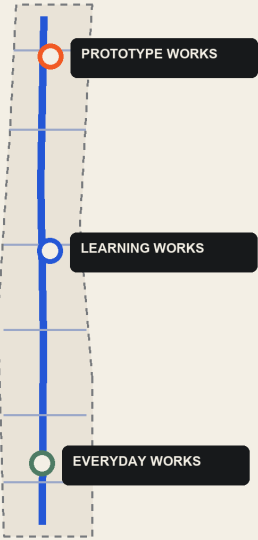
OVERALL
Land, mobility, public service

RETURN
Problems return to R&D and maintenance

JING-ZHANG WORKS

HOW ONE TASK GETS DONE

A problem moves through Prototype, Learning, and Everyday Works; failure returns through the second service chain

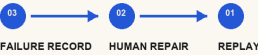


TWO SERVICE CHAINS

LAB TO STREET



PROBLEM TO CHANGE



ANY FAILED STEP REACHES A PERSON AND PAPER

LEARN METHODS FROM SIX CASES, NOT THEIR LOOK

Shared platforms, continuous feedback, agile pilots, resident validation, liveability metrics, and evaluation before scale are rewritten as Jing-Zhang adoption rules. No case imagery or brand asset is copied.



Qinghe Railway Station West Square; arrival, crossing, and station-city interface reference
PHOTO-09 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



Accessible ramp at Tuqiao Station; a continuous-ramp reference from another Beijing location
PHOTO-14 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



Guidance sign in Beijing Railway Station; multilingual and continuous-wayfinding reference
PHOTO-15 · Siyuwj · CC BY-SA 3.0 · External context reference; not field evidence

SHARED
Punggol

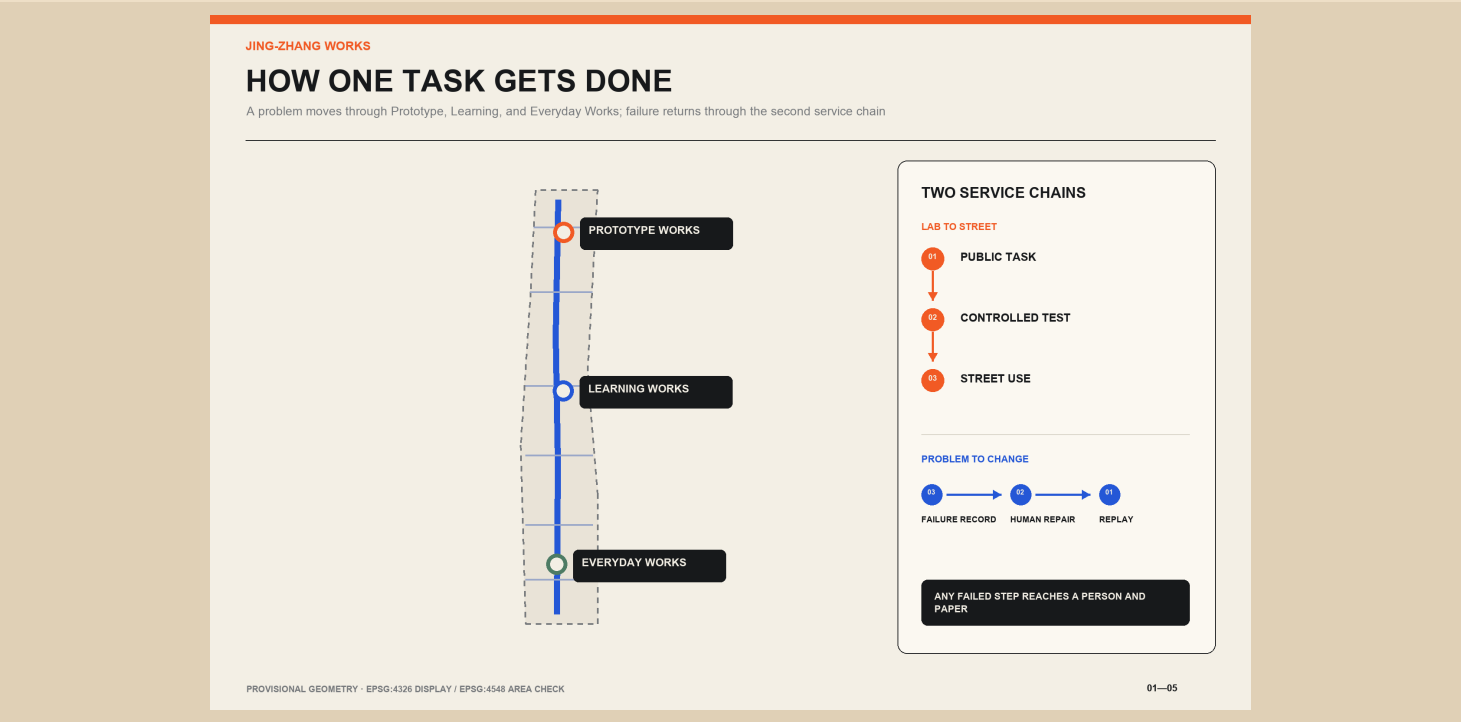
AGILE
Kalasatama and Paris

REAL USE
Woven City and Magok

PUBLIC RIGHT
Barcelona liveability

THREE WORKS, TWO CHAINS THAT CAN TURN BACK

Lab to Street controls release. Problem to Change controls maintenance, suspension, and retirement. Technology must be able to return when it fails.



PROTOTYPE
Controlled test before street

LEARNING
Lower service barriers

EVERYDAY
Face commute and daily life

WINGS
Tech service and scenarios



Locomotive display in Jingzhang Railway Park; industrial-culture and test-space context
PHOTO-06 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



Student reading room at the University of Science and Technology Beijing; shared-learning and night-study reference
PHOTO-16 · Ccyber5 · Public domain dedication by the copyright holder · External context reference; not field evidence



North concourse of Line 13 Dazhongsi Station; everyday transit and staffed-service interface reference
PHOTO-11 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence

PROTOTYPE WORKS: CONTAIN FAILURE FIRST

An observation foyer, accessible test lane, isolated test pocket, physical stop, and failure review make a controlled section. A robot that fails here never reaches the public street.

COMPONENT C04 accessible test lane	OWNER Test lead and site safety
RESOURCE M-class movable kit	HARD STOP Branch approach or failed E-stop

JING ZHANG WORKS

THREE WORKS, THREE SPATIAL PROTOTYPES

One service rule becomes controlled testing, public learning, and everyday service space

01 PROTOTYPE WORKS

02 LEARNING WORKS

03 EVERYDAY WORKS

PROVISIONAL GEOMETRY · SPSS-A330 DISPLAY · SPSS-A340 AREA CHECK

01-05

Locomotive display in Jingzhang Railway Park; industrial-culture and test-space context
PHOTO-06 · N599FZ · CC-BY-SA 4.0 · External context reference; not field evidence

PROVISIONAL GEOMETRY · FIELD PERFORMANCE UNKNOWN

official_boundary=false · concept only

LEARNING WORKS: MAKE THE NEXT STEP CLEAR

Public arrival, Ask First, a shared table, paper material, and human sign-off form a service-and-learning section. A visitor without a smartphone still knows whom to ask.

COMPONENT	OWNER
C01 Ask First + C05 marker	Service desk and venue
RESOURCE	HARD STOP
S-class map, signs, staff	Stale source or absent staff



Guidance sign in Beijing Railway Station; multilingual and continuous-wayfinding reference
PHOTO-15 · Siyuwj · CC BY-SA 3.0 · External context reference; not field evidence



Student study room at the University of Science and Technology, Beijing; shared-learning and night-study reference
PHOTO-16 · Siyuwj · CC BY-SA 3.0 · External context reference; not field evidence

EVERYDAY WORKS: SURVIVE AN ORDINARY EVENING

A shaded entry, quiet rest, human takeover, and public change record form an everyday section. Dazhongsi siting remains unverified, so every component stays movable.

COMPONENT
C02 Takeover + C03 Change Log

OWNER
Safety lead and service owner

RESOURCE
S/M movable kit

HARD STOP
Crowd, error, or blocked exit



Accessible lift at Dazhongsi Station Exit E; step-free arrival reference
PHOTO-12 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence

North concourse of Line 13 Dazhongsi Station; everyday transit and staffed-service interface reference
PHOTO-05 · N0007Z · CC BY-SA 4.0 · External context reference; not field evidence

NOT PERSONAS: EIGHT MOMENTS WHERE A TASK BREAKS

No phone, wheelchair, low vision, care work, neighbourhood life, study, testing, and visiting from abroad each map to a complete, walkable task—not an abstract avatar.



Accessible elevator at Dajiaoting Station; a service-facility reference from another Beijing location
PHOTO-13 · Wcr1993 · CC BY-SA 4.0 · External context reference; not field evidence



Accessible ramp at Tuqiao Station; a continuous-ramp reference from another Beijing location
PHOTO-14 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



Guidance sign in Beijing Railway Station; multilingual and continuous-wayfinding reference
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Student reading room at the University of Science and Technology Beijing; shared-learning and night-study reference
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EIGHT PUBLIC TASKS, EACH WITH A NO-APP ROUTE

Routes, language, shade, night study, older-adult service, sensory access, jobs, and crowd passage share one floor: a person can find staff and leave with paper.



Qinghe Railway Station West Square; arrival, crossing, and station-city interface reference
PHOTO-09 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



South underground corridor of Qinghe Railway Station; continuous wayfinding and walking-interface reference
PHOTO-10 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



Accessible lift at Dazhongsi Station Exit E; step-free arrival reference
PHOTO-12 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence

MOBILITY
Arrival, shade, crowd

SERVICE
Older, language, sensory access

GROWTH
Night study, jobs, events

COMMON RULE
Same task without an app

FOUR INDUSTRY TESTS THAT FIRST LEARN TO STOP

Robots, closed delivery, privacy sensing, and digital twins stay controlled. Boundary breach, human proximity, unclear data purpose, or failed command triggers a hard stop.

ROBOT
Parallel access and recovery
DELIVERY
Closed-zone speed and takeover
SENSOR
Aggregate and delete

TWIN
Synthetic data and human command

Locomotive display in Jingzhang Railway Park; industrial-culture and test-space context
PHOTO-06 · N509FZ · CC-BY-SA 4.0 · External context reference; not field evidence

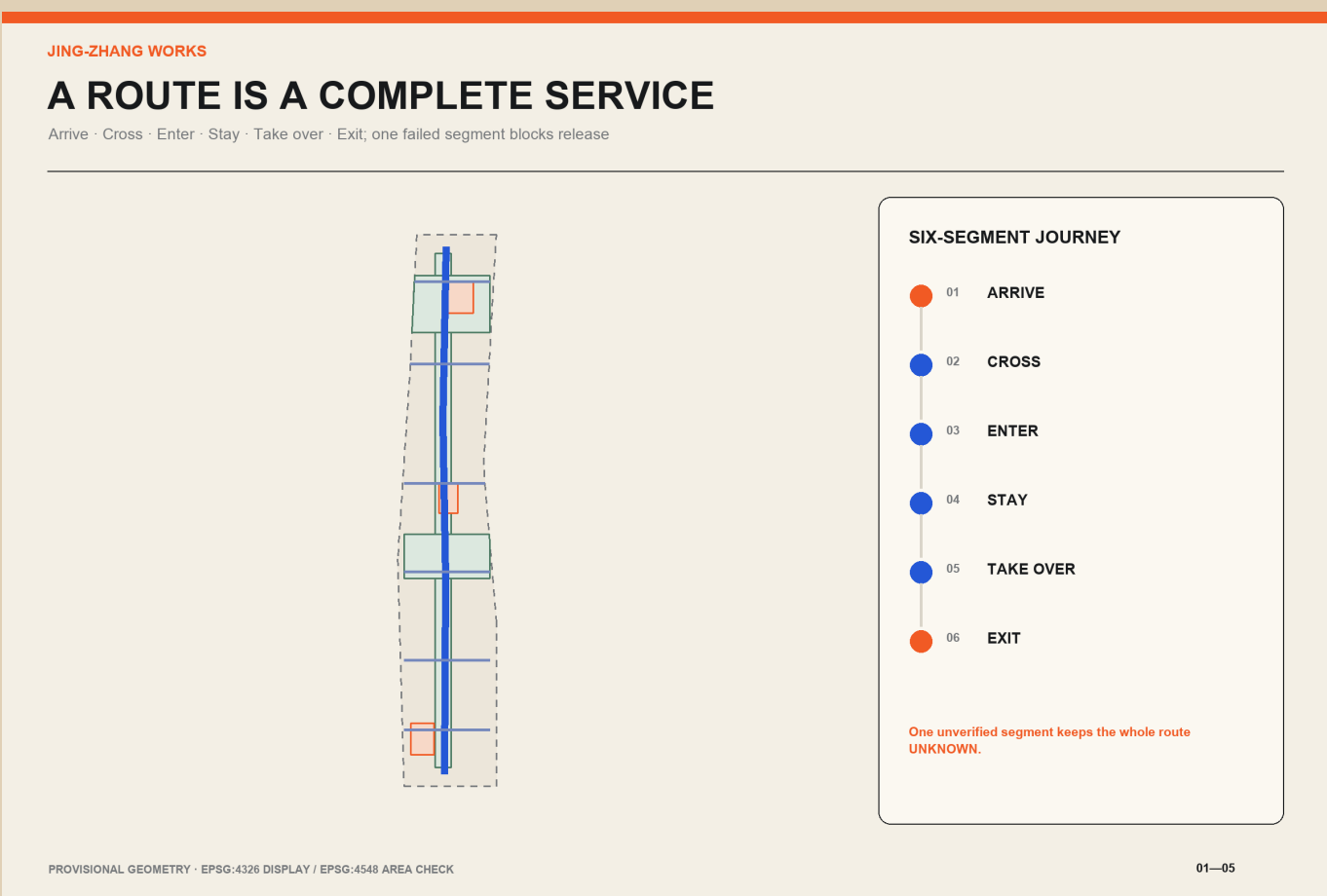
PROVISIONAL GEOMETRY · FIELD PERFORMANCE UNKNOWN

South portal of the Badaling tunnel, 1909 or earlier; engineering-history context
PHOTO-02 · 1876-1915 · Public domain in China · External context reference; not field evidence

official_boundary=false · concept only

ONE ROUTE COUNTS FROM ARRIVAL TO EXIT

Arrival, crossing, entry, stay, takeover, and exit form one service. One unverified segment blocks the route; a polished fragment never cancels a broken middle.



South underground corridor of Qinghe Railway Station; continuous wayfinding and walking-interface reference
PHOTO-10 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence

01
ARRIVE

02
CROSS

03
ENTER

04
STAY

05
TAKEOVER

06
EXIT

LAND EXPLAINS RELATIONSHIPS; BUILDINGS KEEP THEIR UNKNOWN

Six concept zones check topology and task relationships inside the provisional frame. FAR, height, ownership, structural safety, and building action wait for professional evidence.

JING-ZHANG WORKS

PROGRAMME FOLLOWS PUBLIC TASKS

Six concept zones cover the provisional frame; statutory use and intensity stay unknown

Research and development testing

Qinghe and continuous green space

Learning and innovation services

Industry and public services

Community and cultural services

EIGHT-RESOURCE ORDER

01 LAND

02 SPACE

03 INDUSTRY

04 FUNDING

05 TALENT

06 COMPUTE

07 DATA

08 SCENARIOS

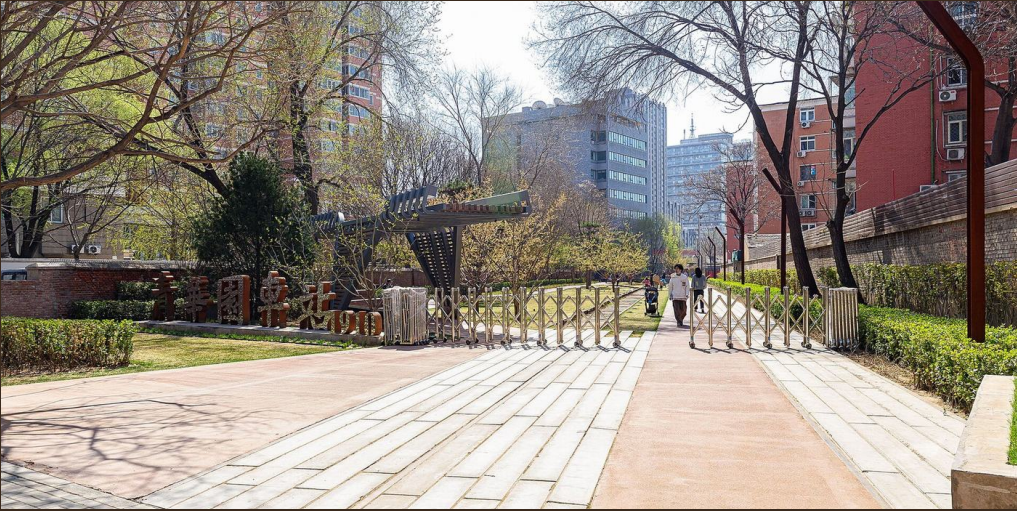
A resource is unsecured until authorization, quantity, term, and exit are recorded.

PROVISIONAL GEOMETRY · EPSG:4326 DISPLAY / EPSG:4548 AREA CHECK

01—05



Old Qinghuayuan Station after renovation; adaptive-reuse reference
PHOTO-04 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence



North entrance of Old Qinghuayuan Station Park; railway heritage entering daily public space
PHOTO-05 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence

LAND	RESOURCES	KNOWN	UNKNOWN
Six-zone topology coverage 1.0	Eight-item adoption order	Concept layers and footprints	Statutory use, FAR, height, building action

NO SMARTPHONE, ONE COMPLETE TASK

U05-A is not a result photograph. It is an unexecuted receipt: arrive, Ask First, read the checklist, obtain human sign-off, recheck quietly, and leave with paper. Field completion remains zero.

WORK ORDER · U05-A · NOT EXECUTED

01

Arrive at numbered entry

02

State need at Ask First

03

Read dated paper checklist

04

Human checks and signs

05

Recheck in a quiet seat

06

Leave safely with paper



Guidance sign in Beijing Railway Station; multilingual and continuous-wayfinding reference
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South underground corridor of Qinghe Railway Station; continuous wayfinding and walking-interface reference
PHOTO-10 · N509FZ · CC BY-SA 4.0 · External context reference; not field evidence

12/12 RULES COMPLETE, 0/12 FIELD TASKS COMPLETE

Structural completeness does not mean the city works. Only a replay by a real user, operator, and qualified reviewer can turn unknown into a measured value.



Qinglongqiao Railway Station in 1909; Jing-Zhang historical context
PHOTO-01 · Public domain in China · External context reference; not field evidence

12/12
STRUCTURAL CONTRACTS READY

0/12
AUTHENTIC FIELD TASKS COMPLETE

Zhongguancun Plaza and street at dusk; public-life context in the innovation district
PHOTO-08 · Charlie fong · Public domain dedication by the copyright holder · External context reference; not field evidence